

Switch Class Documentation

Source File: Port.h
Namespace: cap
Class Header: class Port : public Dev

Overview

The *Port* class is a *Dev* that accepts a *Word* input whose output is a selected bit of the *Word*.

Constructors

- Port() (default constructor)
 - **Purpose:** Sets the word to null.
- Port(const Port& obj) (copy constructor) [deleted]

Destructor

- ~Port() [virtual]
 - **Purpose:** Removes the word.

Assignment Operator

- operator=(const Port& rhs) [deleted]

Member Functions

- bind(Dev& obj) [private]
- unbind() [private]
- input(Word& obj)
 - **Purpose:** Assigns *obj* as the word and makes the selected bit the most significant bit, and then returns true if no word was assigned; otherwise, it returns false.
 - **Parameters:**
 - *obj*: *Word* reference object.
 - **Return:** Boolean.
- set(size_t idx)
 - **Purpose:** Makes the selected bit *idx* if word is not null and *idx* is less than the size of the word.
 - **Parameters:**
 - *idx*: Unsigned integer.
- valid() const [final]
 - **Purpose:** Returns true if it has a word input; otherwise, it returns false.
 - **Return:** Boolean.
- output() const [final]
 - **Purpose:** Returns the value of the selected bit of its word if it is valid; otherwise, it returns false.
 - **Return:** Boolean.
- toString() const [final]
 - **Purpose:** Returns "T" if the selected bit of its word is true, "F" if the selected bit of its word is false, or "X" if it is invalid.
 - **Return:** String.